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A formation maintenance and reconstruction method of UAV swarm based on distributed control

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ABSTRACT

This paper studies the problem of formation maintenance and reconstruction of UAV swarm without obstacles firstly. By designing the position and speed consistency control law between UAVs, each UAV and the virtual leader, the swarm formation control and maintenance is realized. By changing the relative positional relationship between each UAV and the virtual leader, the swarm formation transformation is realized. Secondly, the problem of formation maintenance and reconstruction of UAV swarm with obstacle avoidance is studied. A collision prediction mechanism is introduced to determine whether each UAV needs to avoid obstacles or not. Then, on the basis of the above control law, the artificial potential field method is combined to realize the swarm formation control and maintenance with obstacle avoidance. The disadvantage of the swarm formation transformation method is the position of each UAV in the target formation is preset, so CBBA is used to allocate the positions from the initial formation to the target formation to realize flexible and effective formation transformation. The simulation results show that the UAV swarm can generate, maintain and reconstruct the desired formation in a real-time distributed manner while avoiding obstacles effectively.

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1. Introduction

Formation control is the premise to ensure UAV swarm performing various tasks smoothly, and has become a new research hotspot in academia. UAV swarm formation control refers to multi-UAV with independent capacity to generate, maintain and reconstruct corresponding formations according to different mission requirements or realistic environment. Different swarm formations have different uses, and common formations are V-shape, straight-shape, circle-shape and so on. V-shape can reduce fuel consumption and increase flight range by means of upwash airflow, straight-shape is used to go through obstacles like narrow valleys, and circle-shape is used to attack targets cooperatively.

The solution strategy of UAV swarm formation control is mainly divided into two categories: centralized method and distributed method. The centralized method relies on valid communication between UAVs and the central node, once the central node disabled, it will lead to a failure. Compared with the centralized method, the distributed method pays more attention to the mutual autonomy coordination among UAV swarm, and has the characteristics of decentralization, automation and autonomy [1,2]. Among them,

the formation control based on Consensus Theory is a typical distributed method and is widely used in this field [3–8]. Reynolds presented the famous Boid model, which requires each agent to satisfy the three rules of separation, aggregation and speed matching to maintain close clustering and achieve consistent speed [3]. Based on this, Olfati realized the swarm control of multi-agent through the artificial potential function and speed consistency, while there is a disadvantage that the swarm can only form a α crystal structure [4,5]. To solve this problem, the leader-follower algorithm achieves the maintenance of various formations by specifying the leader's trajectory and keeping the specific geometric positional relationship between UAVs and the leader in the space [6]. However, this method is very dependent on the leader. Once the leader has some problems, the swarm will fail. Then, the virtual structure method was proposed. The virtual structure method specifies a virtual point as a reference point, the position of each agent is determined by this virtual reference point to avoid the above problem [7,8].

Rapid and efficient formation transformation is a key factor to ensure UAV swarm performing tasks successfully. Most of the UAV swarm formation transformation control methods based on Consensus Theory need to specify the target position of each UAV from the initial formation to the target formation in advance [9]. As a result, the global optimization during formation transformation cannot be satisfied, the real-time requirement cannot be met as well.

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Sui Zezhi presented a nested optimization strategy based on Hungarian algorithm and Particle Swarm Optimization (PSO) to achieve rapid assignment of UAVs' target positions [10]. However, as a centralized task assignment strategy, Hungarian algorithm cannot be applied to distributed UAV swarm formation control well. The Consensus-Based Bundle Algorithm (CBBA) is a distributed auction algorithm for solving multi-agent multi-task assignment problem. Each agent negotiates by autonomous competing to achieve a consistent distribution result without the control of the central node [11].

In this paper, we first design a cooperative control law based on the consensus algorithm to realize formation maintenance and reconstruction of UAV swarm. Then, we present a collision prediction mechanism to determine whether each UAV needs to avoid the detected obstacles or not. And the artificial potential field is combined with the consensus algorithm to realize the swarm formation maintenance and reconstruction with obstacle avoidance. Finally, we treat the total distance cost as the optimization goal, and propose an optimization strategy of the swarm formation transformation based on CBBA, which is the most important novelty of this paper. Using CBBA to allocate the position of each UAV from the initial formation to the target formation, the control law designed in this paper can well suited to the formation transformation problem of UAV swarm in a real-time distributed manner.

2. A formation maintenance and reconstruction method based on the consistency

2.1. UAV swarm cooperative control

The dynamic model of the UAV can be expressed as

$$\begin{bmatrix} \dot{V}_i \\ \dot{\psi}_i \\ \dot{\varphi}_i \\ \dot{X}_i \\ \dot{Y}_i \\ \dot{Z}_i \end{bmatrix} = \begin{bmatrix} -g \sin \psi_i \\ -g \cos \psi_i / V_i \\ 0 \\ V_i \cos \psi_i \cos \varphi_i \\ V_i \cos \psi_i \sin \varphi_i \\ V_i \sin \psi_i \end{bmatrix} + \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1/V_i & 0 \\ 0 & 0 & 1/(V_i \cos \psi_i) \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} a_{1i} \\ a_{2i} \\ a_{3i} \end{bmatrix} \quad (1)$$

where V_i is the scalar velocity of UAV i , ψ_i is the pitch angle of UAV i , φ_i is the heading angle of UAV i , X_i, Y_i, Z_i are the coordinates of UAV i in three-dimension space. a_{1i} is the tangential acceleration of UAV i , a_{2i}, a_{3i} are normal acceleration components in vertical and horizontal directions of UAV i . Linearize the dynamic model of UAV i and we can get a new control input which has a relationship with the old control input as follows

$$\begin{cases} u_{xi} = a_{1i} \cos \psi_i \cos \varphi_i - a_{2i} \sin \psi_i \cos \varphi_i - a_{3i} \sin \varphi_i \\ u_{yi} = a_{1i} \cos \psi_i \sin \varphi_i - a_{2i} \sin \psi_i \sin \varphi_i + a_{3i} \cos \varphi_i \\ u_{zi} = a_{1i} \sin \psi_i + a_{2i} \cos \psi_i \sin \varphi_i - g \end{cases} \quad (2)$$

Here, we assume that all UAVs are flying at the same altitude and simplify the model into the two-dimension space. Thus, the second-order motion model of the swarm can be expressed as

$$\begin{cases} \dot{x}_i(t) = v_i(t) \\ \dot{v}_i(t) = u_i(t) \end{cases} \quad i = 1, 2, \dots, N \quad (3)$$

where $x_i(t)$ is the position vector of UAV i ; $v_i(t)$ is the speed vector of UAV i , $|v_i(t)| \leq v_{\max}$; $u_i(t)$ is the control input vector of UAV i , and N is the number of UAVs in the swarm.

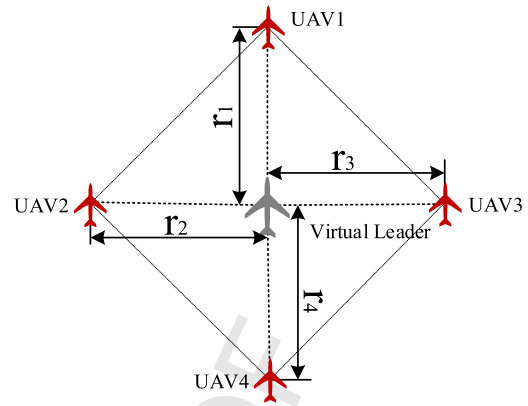


Fig. 1. The formation in steady state of UAV swarm.

And we assume that each UAV cannot change its velocity immediately because of its mass and aerodynamic effects, and there is a maximal acceleration $a_{\max}$ for each UAV.

The communication topology relationship between UAVs can be represented by a directed graph $G = (V, E)$. Where $V = \{1, 2, \dots, N\}$ represents the set of UAVs, and $E = \{(i, j) | i, j \in V, i \neq j\}$ represents the set of communication edge between UAVs. $A = [a_{ij}]$ is the adjacency matrix of this undirected graph. $a_{ij} = 1$ when $(i, j) \in E$, otherwise, $a_{ij} = 0$.

The communication range of each UAV is finite. Assume the communication range of the UAV is r_c . Once the distance of the two UAVs is larger than r_c , they will not be able to communicate with each other. Only within this range can information be transmitted between UAVs.

The formation maintenance and reconstruction method based on consistency refers to designing the cooperative control law of the UAV swarm based on consensus theory to control the swarm realizing the formation maintenance and formation transformation during the flight. Define a virtual leader as the geometric reference point of the swarm, $x_v(t)$ is the virtual leader's position vector and $v_v(t)$ is the virtual leader's speed vector. The position vector of UAV i relative to the swarm geometry reference point is r_i . UAV i only needs to exchange its position and speed information with the UAVs in its communication neighborhood, so that the speed of each UAV tends to be the same, and each UAV reaches the desired distance from the geometric reference point, thereby ensuring the generation and maintenance of the swarm formation. That is to say, $\lim_{t \rightarrow \infty} (v_i(t) - v_j(t)) = 0$ and $\lim_{t \rightarrow \infty} (x_i(t) - x_v(t)) = r_i$.

The formation in steady state is shown in Fig. 1.

In the process of formation transformation, the UAV swarm only needs to change the relative position vector r_i of each UAV according to the desired formation shape. That is, the swarm needs to determine each UAV's relative position in the desired formation in advance. For example, in Fig. 2, each UAV goes to its target position in the desired V-shape formation from the initial random state according to the same ID principle (i.e. UAV 1 goes to the position 1). When converting from V-shape to straight-shape, the UAV will find its relative target position in the straight-shape formation by the same way. By controlling the r_i of each UAV changing from the preset value in V-shape to the preset value in straight-shape, the transformation of the formation is achieved.

This paper assumes that there is a virtual leader in the UAV swarm flying along a given trajectory. All UAVs in the swarm have the capability to detect obstacles and can obtain the position and velocity of the virtual leader. The r_i of each UAV is preset according to the desired formation. Based on the Consensus Algorithm, the cooperative control law is designed as follows:

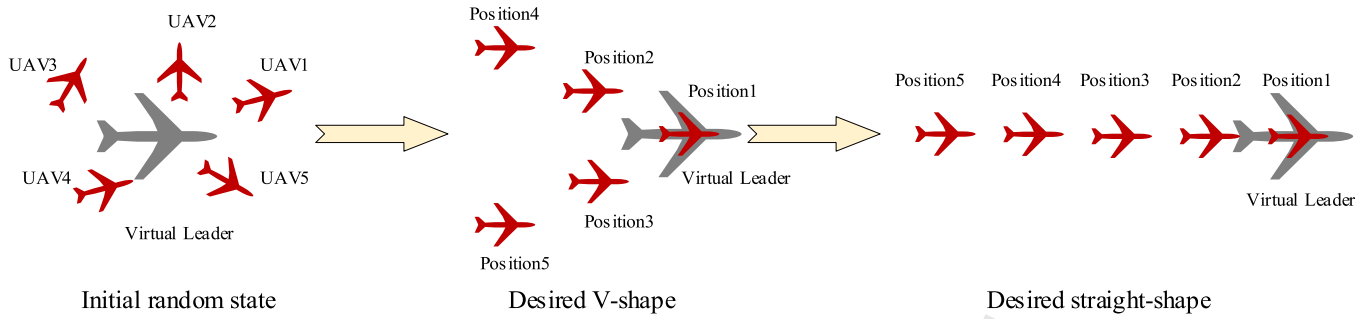


Fig. 2. Formation transformation process.

$$u_i = u_\gamma - k_1 \sum_{j \in N_{\alpha i}} (x_i(t) - x_j(t) - r_{ij}) - k_2 (x_i(t) - x_\gamma(t) - r_i) - k_3 \sum_{j \in N_{\alpha i}} (v_i(t) - v_j(t)) - k_4 (v_i(t) - v_\gamma(t)) \quad (4)$$

where u_γ is the acceleration of the virtual leader, r_{ij} is the expected relative position vector between UAV i and UAV j , and $r_{ij} = r_i - r_j$. k_1, k_2, k_3, k_4 are corresponding gain factors. $N_{\alpha i}$ represents the communication neighborhood which is specifically defined as follows:

$$N_{\alpha i} = \{ \|x_j - x_i\| < r | i, j \in N \} \quad (5)$$

2.2. Simulation results

Assume there are 9 UAVs for swarm flight, the initial speed of each UAV is randomly generated in $[0, v_{\max}] \times [0, v_{\max}]$, and the initial position of each UAV is randomly generated in $[0, 300] \times [0, 300]$. There is a virtual leader in the swarm with initial position $x_\gamma(0) = (150, 150)$ and speed $v_\gamma(0) = (20, 20)$, and the acceleration $u_\gamma = 0$. During the swarm flight, the specific parameters are set as follows: UAV's communication radius $r = 300$, the expected distance d between UAVs is 200. The maximum speed of the UAV is 60, and the maximum acceleration of the UAV is 10. $k_1 = k_2 = k_3 = 1, k_4 = 2$, and the step-size is 0.03. The expected formation is circle-shape. The virtual leader is located at the center of the circle, and other UAVs are evenly distributed on the circumference. The calculation of each UAV's r_i in the circle-shape formation is as follows:

$$R = d / (2 \sin(\pi / N)) \quad (6)$$

$$r_i = (R \sin(360i / N), R \cos(360i / N)) \quad i = 1, 2, \dots, N \quad (7)$$

where, R is the radius of the circle.

The simulation results under control input (4) are shown in Fig. 3.

As is shown in Fig. 3, blue circles represent UAVs, black star represents the virtual leader, the length and the direction of the red arrow respectively represents the speed and the heading of each UAV, and blue line represents the communication relationship between UAVs. It can be seen that the UAV swarm eventually forms the expected formation.

Fig. 4 shows the error between the average position (x-axis and y-axis) of all UAVs and the position of the virtual leader at each moment. It can directly reflect the error variation of the swarm formation. It can be seen that the position error finally converges to zero and the swarm formation is stable.

Fig. 5 shows the error between the average speed (x-axis and y-axis) of all UAVs and the speed of the virtual leader at each

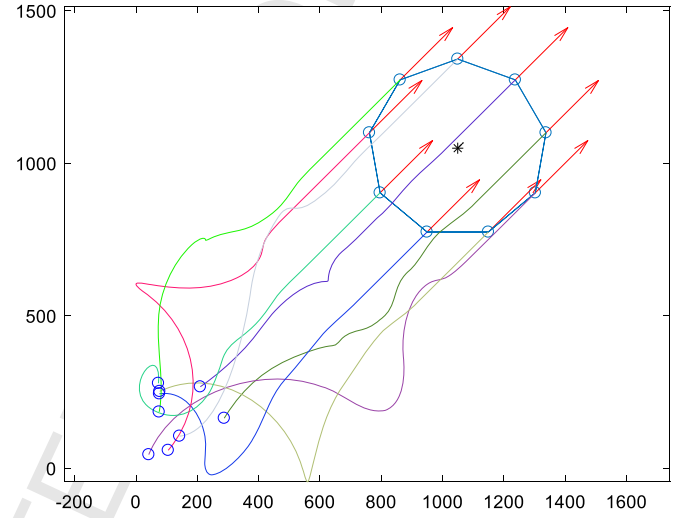


Fig. 3. The swarm trajectory. (For interpretation of the colors in the figure(s), the reader is referred to the web version of this article.)

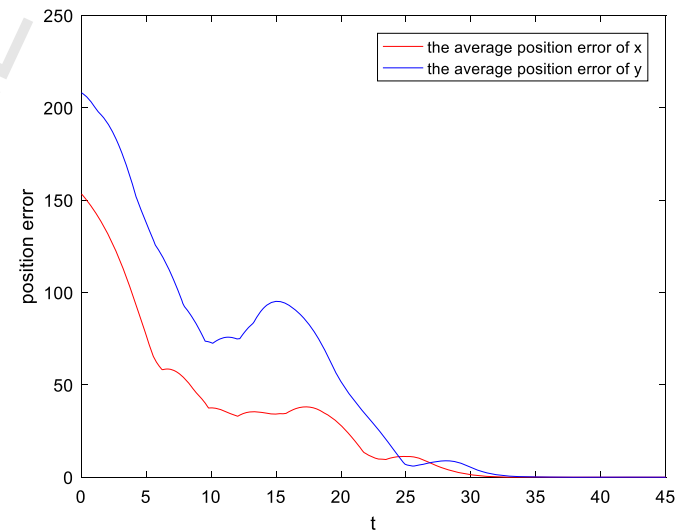


Fig. 4. Position error between UAVs and the virtual leader.

moment. It can directly reflect the error variation between the expected speed of the swarm and the average speed of the current swarm. It can be seen that the velocity error finally converges to zero and the swarm speed is stable.

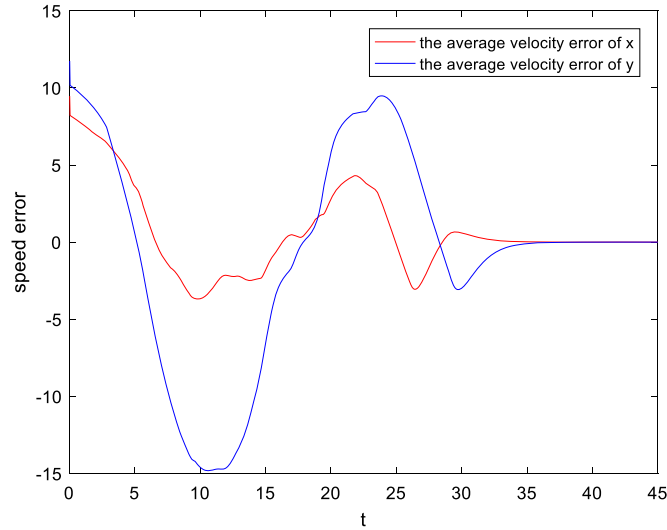


Fig. 5. Speed error between UAVs and the virtual leader.

3. A formation maintenance and reconstruction method with obstacle avoidance

After detecting the presence of obstacles, UAV swarm should change its formation to avoid the obstacles, and then reconstruct its formation after the obstacle avoidance is completed. The formation control law (4) based on the consensus algorithm in the previous section can only ensure that the swarm moves along the direction of the virtual leader, and cannot adjust its direction according to the realistic situation to avoid obstacles. Therefore, a method combining with the obstacle avoidance mechanism based on the artificial potential field method is designed to ensure the maintenance and reconstruction of the swarm formation during avoiding obstacles.

The artificial potential field method avoids obstacles by introducing mechanism of virtual forces between UAVs and between each UAV and each obstacle. There are attractive and repulsive fields between UAVs, while there is only a repulsive field between UAVs and obstacles. When two UAVs are close to each other less than the desired distance, there is a repulsive force between them and make them away from each other. When they are away from each other more than the desired distance, there is an attractive force to make them close to each other. And when their distance is exactly equal to the desired distance, the force is zero and the two UAVs reach a steady state. In the same way, when a UAV is close to an obstacle, it will receive a repulsive force from the obstacle to move it away.

3.1. Collision prediction mechanism

Most obstacle avoidance algorithms have a disadvantage that all detected obstacles will generate a repulsive force to the UAV [16]. Aiming at this problem, a new collision prediction mechanism is introduced. After obstacles are detected, each UAV will predict whether it needs to avoid obstacles or not, and avoid the obstacles that will collide with it. It is equivalent to narrowing the obstacles neighborhood of the UAV, so the swarm can ignore the effect of the obstacles that will not collide with it and flight along its predetermined heading, thereby reducing the distance traveled by obstacle avoidance.

Assume there is a static circular obstacle O in the space with radius r_{ok} centered at x_{ok} . The current heading of UAV i is φ when it detects this obstacle. The expected distance between UAV i and the obstacle is d_o . According to UAV i 's current heading, if the

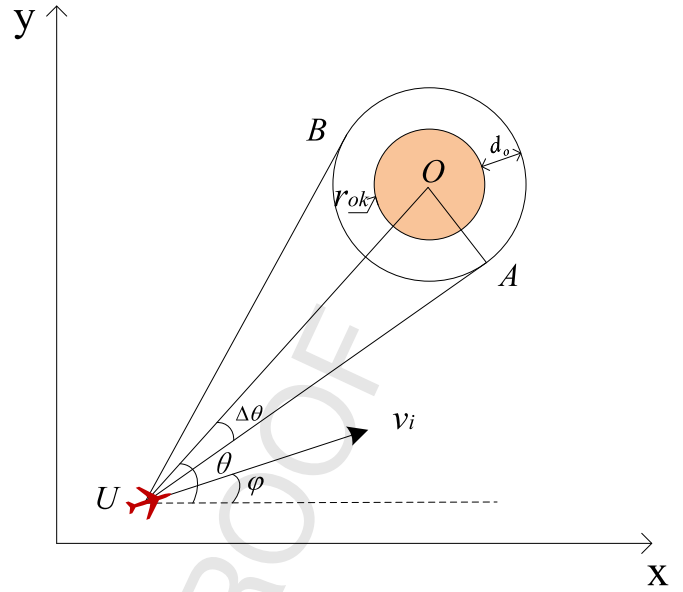


Fig. 6. The collision prediction between the UAV and the obstacle.

velocity vector v_i is in the area UAB in Fig. 6 (the area that is composed of the tangent made from UAV i to the circular O), it is necessary to consider the impact of this obstacle and adjust UAV i 's heading to avoid the obstacle.

The specific mathematical calculation method is as follows:

$$AU = \sqrt{OU^2 - r_{ok}^2} \quad (8)$$

$$\Delta\theta = \arctan \frac{r_{ok}}{AU} \quad (9)$$

If the heading angle of UAV i is in the range of $[\theta - \Delta\theta, \theta + \Delta\theta]$, that is to say

$$\theta - \Delta\theta \leq \varphi \leq \theta + \Delta\theta \quad (10)$$

then, UAV i will be repelled by obstacle O to avoid it.

3.2. Artificial potential function

Define σ norm as a nonnegative map that is differentiable everywhere. The purpose of this norm is to construct a smooth potential function. Its specific definition is in [4]

$$\|z\|_\sigma = \frac{1}{\varepsilon} [\sqrt{1 + \varepsilon \|z\|^2} - 1] \quad (11)$$

and the gradient of σ norm is

$$\nabla \|z\|_\sigma = \frac{z}{\sqrt{1 + \varepsilon \|z\|^2}} \quad (12)$$

There are attractive and repulsive forces between UAVs, and the artificial potential function $\psi_\alpha(z)$ between UAVs is chosen as [4]

$$\psi_\alpha(z) = \int_{d_\alpha}^z \phi_\alpha(s) ds \quad (13)$$

The specific calculation process is as follows:

$$\phi_\alpha(z) = \rho_h(z/r_\alpha) \phi(z - d_\alpha) \quad (14)$$

$$\phi(z) = \frac{1}{2} [(a+b)\sigma_1(z+c) + a-b] \quad (15)$$

$$\sigma_1(z) = \frac{z}{\sqrt{1+z^2}} \quad (16)$$

where $\phi_\alpha(z)$ is an action function between UAVs, and $\phi(z)$ is an uneven sigmoid function. a, b, c satisfies that $0 < a < b$, and $c = |a - b|/\sqrt{4ab}$. r is the communication radius of UAVs, and d is the expected distance between UAVs, $r_\alpha = \|r\|_\sigma$, $d_\alpha = \|d\|_\sigma$.

$\rho_h(z)$ is a continuous smooth bump function from 0 to 1.

$$\rho_h(z) = \begin{cases} 1 & z \in (0, h) \\ \frac{1}{2} \left[1 + \cos(\pi \frac{z-h}{1-h}) \right] & z \in (h, 1), \quad h \in (0, 1) \\ 0 & \text{others} \end{cases} \quad (17)$$

where h is used to determine the spatial area of each UAV's neighborhood that is influenced most by the force field. That is to say, $\rho_h(z) = 1$ when z is in the area of $(0, h)$, and the UAV will be affected by $\phi_\alpha(z)$ most. Once z is beyond $(0, h)$, $\rho_h(z)$ will decrease from 1 to 0, and the effect of $\phi_\alpha(z)$ will decrease to 0 correspondingly. The closer h is to 1, the greater the UAV is affected by other UAVs in its neighborhood.

There is a repulsive force between one UAV and one obstacle, and the artificial potential function $\psi_\beta(z)$ between the UAV and the obstacle is chosen as [4]

$$\psi_\beta(z) = \int_{d_\beta}^z \phi_\beta(s) ds \quad (18)$$

The specific calculation process is as follows:

$$\phi_\beta(z) = \rho_h(z/d_\beta)(\sigma_1(z - d_\beta) - 1) \quad (19)$$

where $\phi_\beta(z)$ is an action function between one UAV and one obstacle, d_β is the expected distance between the UAV and the obstacle and $d_\beta = \|d_\beta\|_\sigma$

3.3. UAV swarm cooperative control

Based on the Consensus Algorithm and the artificial potential field method, the cooperative control law is designed as follows:

$$u_i = u_\gamma + u_{\alpha i} + u_{\beta i} + u_d \quad (20)$$

where u_γ is the acceleration of the virtual leader. $u_{\alpha i}$ is the attraction and repulsion between UAVs. When the distance between UAVs is less than the expected distance d , it appears as repulsive force, and the repulsive force increases with the decrease of d . On the contrary, when the distance is larger than d , it appears as attractive force, and the attractive force increases with the increase of d . $u_{\beta i}$ is the repulsion between UAV i and obstacle that will collide, the closer the UAV is to the obstacle, the larger the repulsive force is received. u_d is the formation maintenance control law of the UAV swarm.

The specific definition is as follows:

$$u_{\alpha i} = -c_\alpha \sum_{j \in N_{\alpha i}} \nabla_{x_i} \psi_\alpha(\|x_j - x_i\|_\sigma) \quad (21)$$

$$u_{\beta i} = -c_\beta \sum_{k \in N_{\beta i}} \nabla_{x_i} \psi_\beta(\|x_{ok} - x_i - r_{ok}\|_\sigma) \quad (22)$$

$$u_d = -k_1 \sum_{j \in N_{\alpha i}} (x_i(t) - x_j(t) - r_{ij}) - k_2(x_i(t) - x_\gamma(t) - r_i) - k_3 \sum_{j \in N_{\alpha i}} b_{ij}(t)(v_i(t) - v_j(t)) - k_4(v_i(t) - v_\gamma(t)) \quad (23)$$

Where c_α represents the weight coefficient of the force between UAVs, c_β represents the weight coefficient of the force between UAV i and obstacle that will collide, and satisfies $c_\alpha < c_\beta$.

$a_{ij}(t) = \rho_h(\|x_j - x_i\|_\sigma / \|r\|_\sigma)$ is the element of adjacency matrix A . k_1, k_2, k_3, k_4 are corresponding gain factors. N_o represents the set of all obstacles, $N_{\beta i}$ represents the obstacle neighborhood of UAV i which is specifically defined as follows:

$$N_{\beta i} = \{\theta_{ik} - \Delta\theta_{ik} \leq \varphi_i \leq \theta_{ik} + \Delta\theta_{ik} | k \in N_o\} \quad (24)$$

3.4. Simulation results

Assume there are 9 UAVs for swarm flight with obstacles on the way. If the swarm still flies according to the control law designed in section 2, it cannot escape the obstacles. While the control law designed in this section can solve this problem well. Assume the initial position and initial speed of each UAV and the virtual leader in accordance with Section 2. There are two obstacles in the space, obstacle 1 centers at (2000, 2200) with radius of 200, obstacle 2 centers at (2700, 2400) with radius of 180. During the swarm flight, the specific parameters are set as follows: the expected distance d_o between UAVs and obstacles is 200. $a = 1$, $b = 5$, $\varepsilon = 0.1$, $h = 0.2$, $c_\alpha = 1.8$, $c_\beta = 20$, $k_1 = k_2 = 1$, $k_3 = 2.5$, $k_4 = 2$, and other conditions are consistent with the section 2. The expected formation is circle-shape when the swarm flight without obstacles, the expected formation is straight-shape when avoiding obstacles, and the expected formation is V-shape when reconstructing the formation. The simulation results of the swarm flight process are shown in Fig. 7(a)-(h).

From the dynamic simulation results of Fig. 7 (a)-(h), it can be seen that the UAV swarm can form the desired circle-shape formation to fly. For obstacles 1 and 2, collisions are detected, so the straight-shape is transformed to avoid obstacles. Finally, the reconstruction of V-shape formation is completed to keep flight.

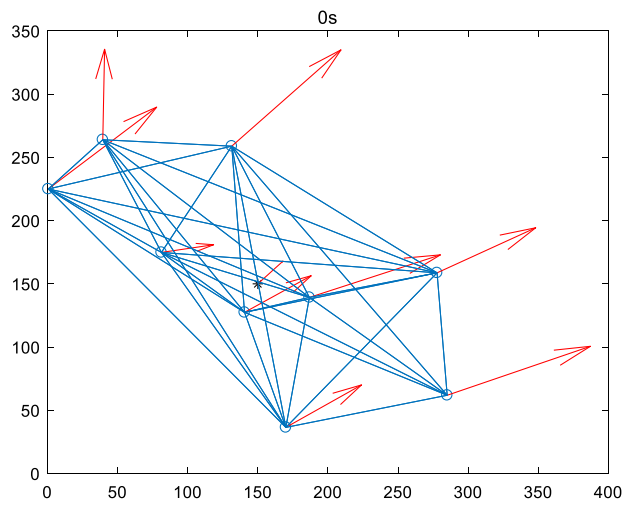
Fig. 8 shows the error between the average position (x-axis and y-axis) of all UAVs and the position of the virtual leader at each moment. Fig. 9 shows the error between the average speed (x-axis and y-axis) of all UAVs and the speed of the virtual leader at each moment. It can be seen that after a series of actions of swarm formation transformation and obstacle avoidance, these two errors tend to 0. The UAV swarm reaches the desired speed and formation shape.

Fig. 10 shows the minimum distance (x-axis and y-axis) between obstacle 1 and each UAV at each moment. Fig. 11 shows the minimum distance (x-axis and y-axis) between obstacle 2 and each UAV at each moment. It can be seen that the minimum distance between UAV swarm and obstacles is always greater than 0, and there is no collision occurs.

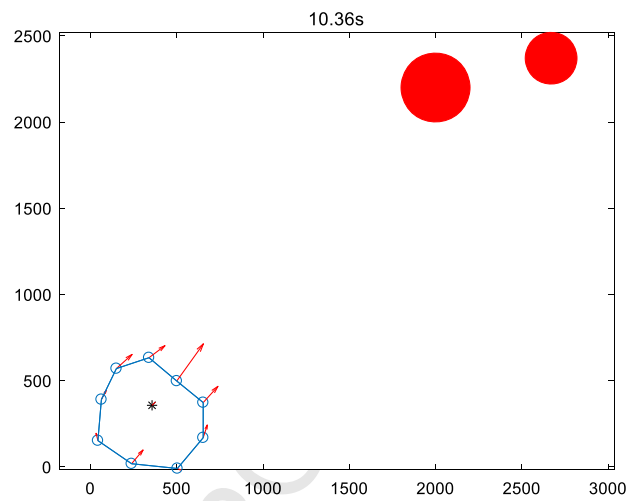
4. Formation maintenance and reconstruction based on CBBA

The UAV swarm formation requires each UAV to maintain a specific positional relationship in space geometrically. In control inputs (4) and (20), the position vector of the UAV relative to the swarm geometry reference point r_i is preset. The disadvantage of this is that the algorithm lacks of flexibility and cannot obtain an optimal allocation solution according to the different situation, thereby increasing the flight cost. In addition, once the UAV is damaged, the above control method cannot adjust the formation according to the number of remaining UAVs during the formation transformation, which will lead to a failure. In this section, CBBA algorithm is used to allocate the positions of the UAV swarm from the initial formation to the target formation to obtain a real-time allocation scheme. Then, according to the swarm cooperative control law designed above, the formation maintenance and reconstruction of the swarm is realized.

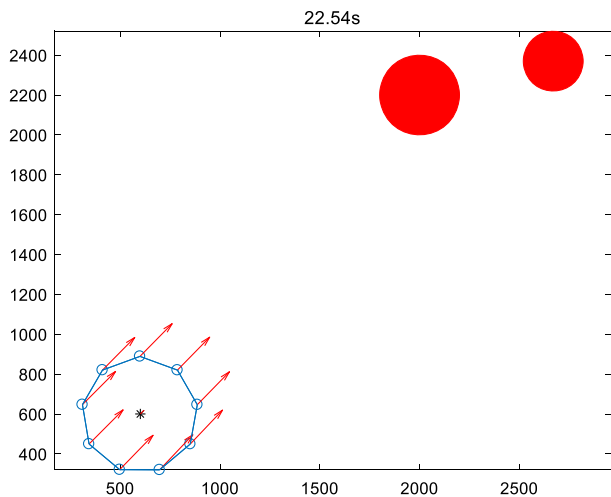
Under the control input (20), the calculation method of the r_i of each UAV is:



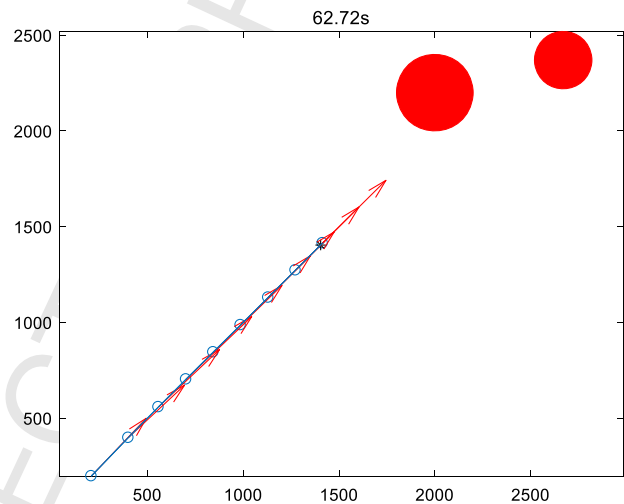
(a) initial state of the swarm



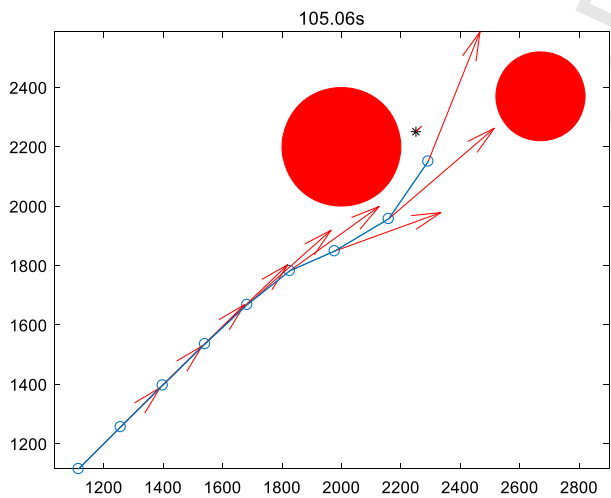
(b) transforming to circle-shape



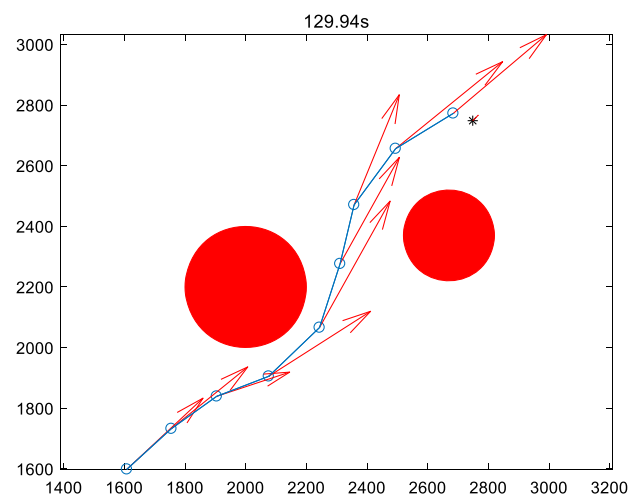
(c) flight with circle-shape



(d) transforming to straight-shape

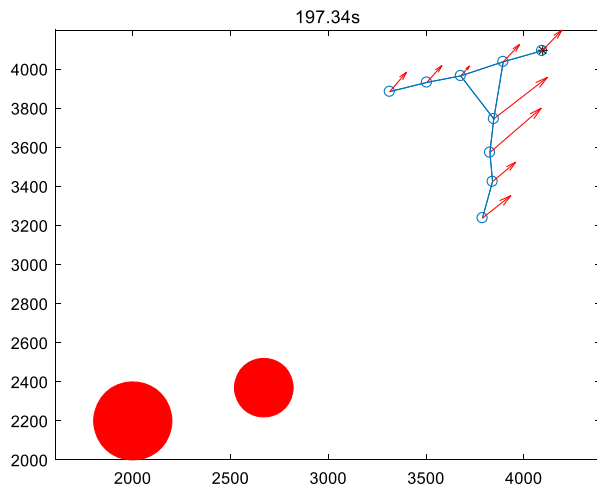


(e) avoiding obstacle 2

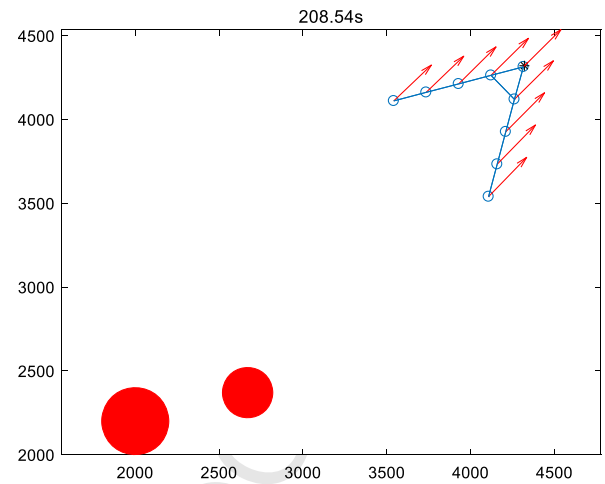


(f) avoiding obstacle 1

Fig. 7. Snapshots of UAV swarm flight.



(g) transforming to V-shape



(h) flight with V-shape

Fig. 7. (continued)

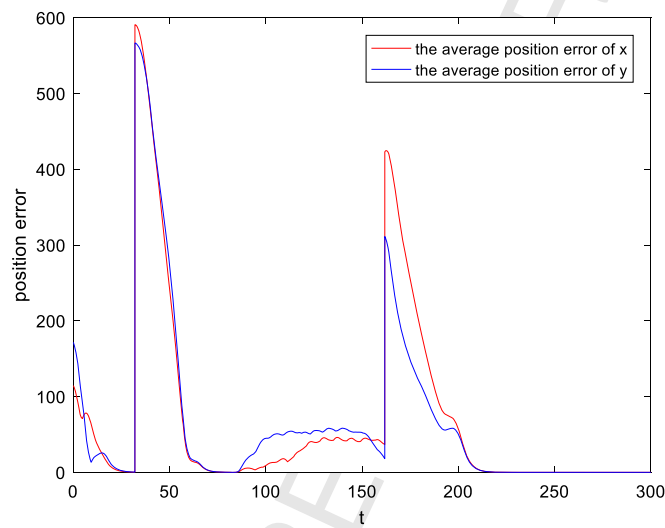


Fig. 8. Position error between UAVs and the virtual leader.

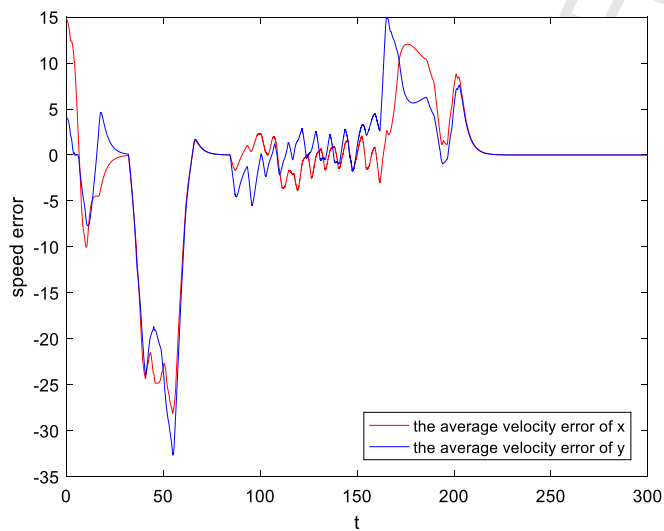


Fig. 9. Speed error between UAVs and the virtual leader.

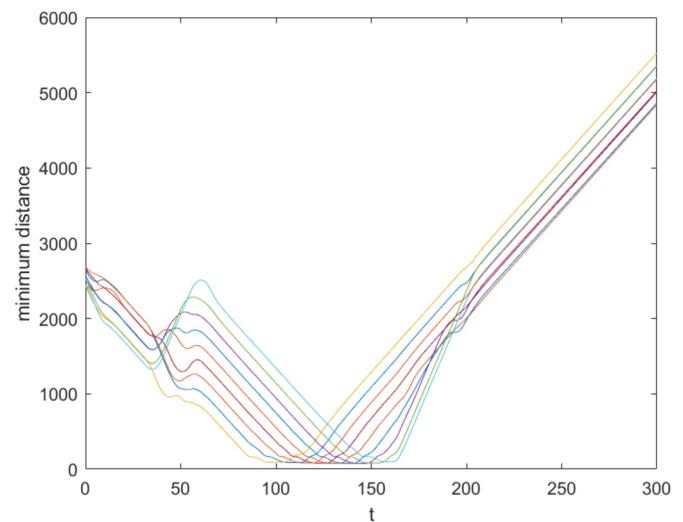


Fig. 10. Minimum distance between obstacle 1 and each UAV.

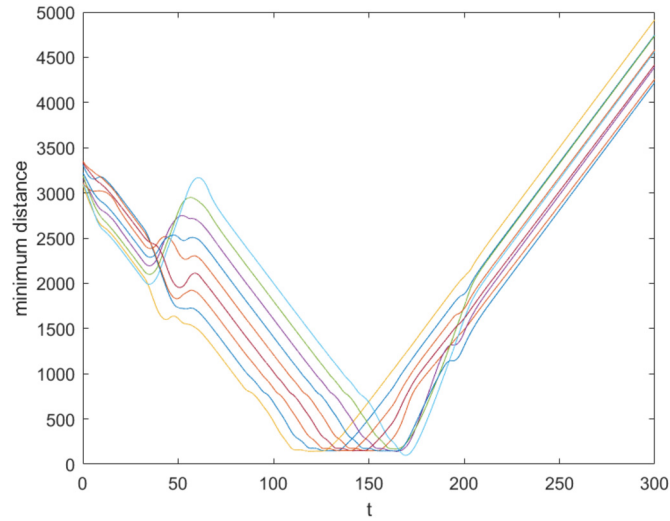


Fig. 11. Minimum distance between obstacle 2 and each UAV.

$$r_i = x_{\gamma} - T_i \quad (25)$$

Where, T_i is the target position of UAV i assigned by CBBA.

4.1. Problem statement of task assignment based on CBBA

Define $T = [T_1^T, T_2^T, \dots, T_N^T]^T$ as the set of N position vectors of the target formation to be assigned to N UAVs. The goal of task allocation is to find a conflict-free task assignment scheme that maximizes the global reward. And the mathematical model of this task assignment problem uses the following distributed integer programming model:

$$\max_{\chi} \sum_{i=1}^N \sum_{j=1}^N c_{ij}(\chi) \chi_{ij} \quad (26)$$

$$\text{s.t.} \quad G(\chi) \leq d$$

$$\chi \in \{0, 1\}^{N \times N}$$

where $\chi \in \{0, 1\}^{N \times N}$ is a set of 0,1 variables in which each decision variable χ_{ij} is used to indicate whether the task j is assigned to UAV i , c_{ij} is the score function for UAV i to perform task j , and $G = g_1, \dots, g_{N_c}^T, d = d_1, \dots, d_{N_c}^T$ is a set of constraint relationships defined in the form of $g_k(\chi, \tau) \leq d_k$. Here, applying to the swarm formation transformation, it means that each UAV can execute one task at most. That is, each UAV can only fly to one of the positions of the target formation.

4.2. Objective function design

This paper takes the total distance cost as the main optimization goal and select the time-discount reward model whose reward will decrease as the task execution time increases. The influencing factors such as the task location, task static reward, UAV position and speed are considered. The reward of task j performed by UAV i is defined as the following form:

$$R_{ij} = R_j e^{-\lambda_j \tau_{ij}} \quad (27)$$

where τ_{ij} represents the time when the UAV i reaches the target position j according to its current speed, λ_j represents the discount factor and R_j represents the reward of the task j . Formation transformation requires all tasks have the same value, and the task reward R_{ij} will decrease based on R_j with the flight time of UAV i .

Here, exponential decay is an indication of the rapid generation of the target formation. According to the flight time τ_{ij} of the UAV i from the current position to the target position, the discount factor λ_j is used to reduce the task reward R_{ij} accordingly. For a given UAV i , the objective function is defined as follows:

$$c_{ij} = R_{ij} \chi_{ij} \quad (28)$$

4.3. CBBA

CBBA is a distributed auction algorithm for solving multi-agent multi-task assignment problem. It is mainly divided into two phases: bundle building phase and Consensus phase.

In the bundle building phase, each UAV selects a task independently. The task bundle is constructed in accordance with the greedy principle. The specific steps of the bundle building phase are as follows:

Step 1: Use the objective function to calculate the internal score c_{ij} of the task j in sequence for all tasks that are not in the bundle of UAV i .

Step 2: Compare the score c_{ij} with UAV i 's current winning bid y_{ij} . If UAV i 's bid is larger than the current winner's bid for the same task, then $h_{ij} = 1$, h_{ij} 's formula is as follows:

$$h_{ij} = \begin{cases} 1, & c_{ij} > y_{ij} \\ 0, & c_{ij} \leq y_{ij} \end{cases} \quad (29)$$

Step 3: Find the maximum value of $c_{ij} \cdot h_{ij}$ and its corresponding task j . If $c_{ij^*} \cdot h_{ij^*} > 0$, add the task j to the current bundle.

Step 4: Update the information related to UAV i : winning agent list z_{ij} , winning bid list y_{ij} , and timestamps information s_{ij} .

Once the UAV i completes the construction of its task bundle, it needs to communicate with its neighbors and enter into the consensus phase. It will receive the winning agent list, winning bid list, and timestamps information of other UAVs and share its own information to them. Here, the timestamp s_{ij} represents the time when UAV i receive the latest message of task j from other UAVs. Its update formula is as follows:

$$s_{ik} = \begin{cases} \tau_r, & \text{if } a_{ik} = 1 \\ \max\{s_{mk} | m \in I, a_{im} = 1\}, & \text{otherwise} \end{cases} \quad (30)$$

where τ_r represents the time when UAV i receives the messages from UAV k . The communication among UAVs in the consensus phase is synchronous, so the actual receive time τ_r can be replaced by the current iteration number of the algorithm. Specific synchronous communication decision rules could be found in literature [11].

The algorithm iterates between the bundle building phase and the consensus phase so as to obtain a conflict-free allocation scheme of target positions and realizes the formation transformation rapidly and efficiently with a smaller total distance cost.

4.4. Simulation results

4.4.1. Comparison of the formation transformation efficiency

Assume there are 9 UAVs for swarm flight, the number of UAVs is increased from 9 to 21, and 2 UAVs are added each time. The initial speed of each UAV is randomly generated in $[0, v_{\max}] \times [0, v_{\max}]$, the initial position of each UAV is randomly generated in $[0, 300] \times [0, 300]$. The maximum speed of the UAV is 60, and the maximum acceleration of the UAV is 10. There is a virtual leader in the swarm with initial position $x_{\gamma}(0) = (150, 150)$ and speed $v_{\gamma}(0) = (20, 20)$, and the acceleration $u_{\gamma} = 0$. The expected swarm formation is circle-shape. $\lambda = 0.01$, $R_j = 100$, and the step-size is 0.03. For the two methods of using CBBA to allocate

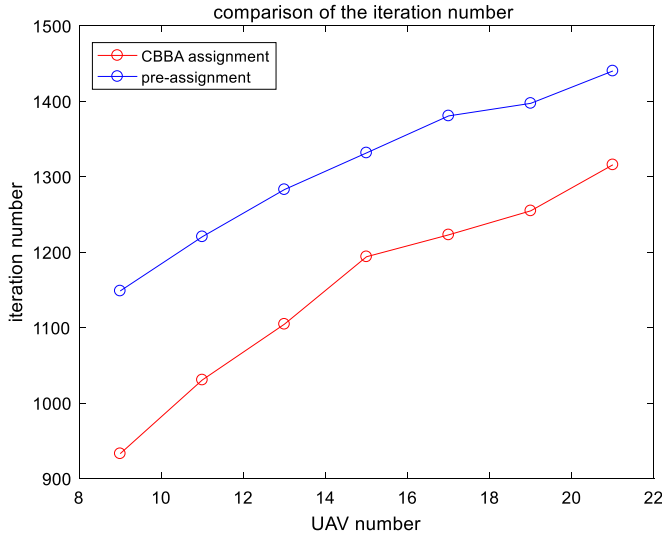


Fig. 12. Comparison of the average iteration number.

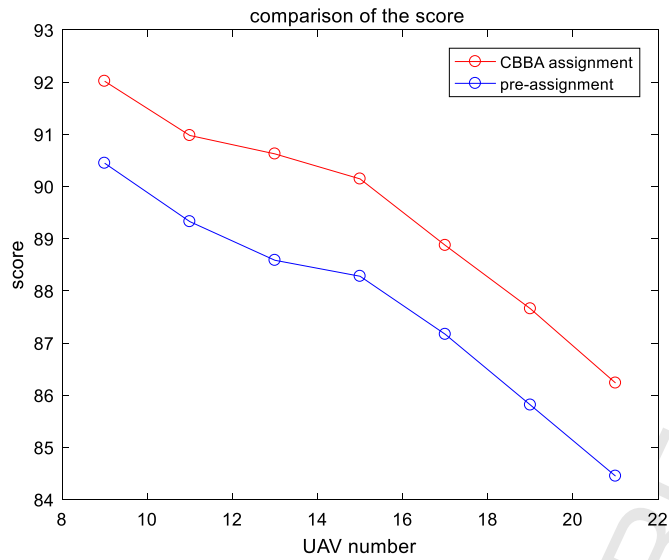


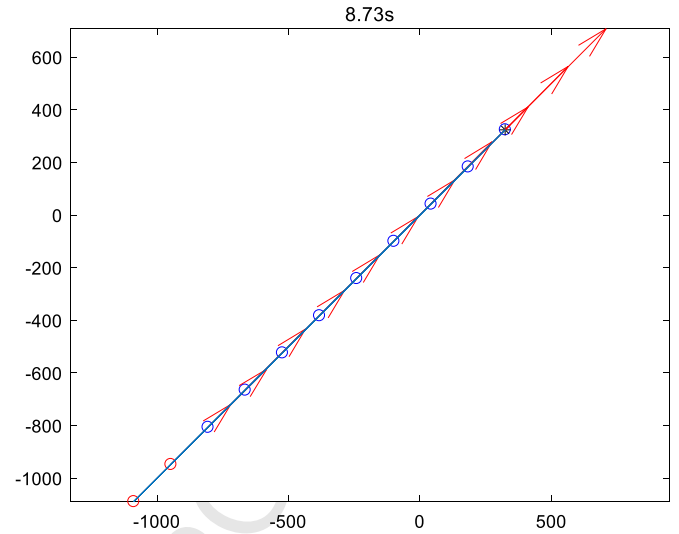
Fig. 13. Comparison of the average score of a single UAV.

target positions and pre-assignment of target positions, 20 comparison experiments are studied for different sizes of UAV swarm. Experiments compare the average iteration number from the initial position to the expected formation and the average score of a single UAV using these two methods respectively. Both methods use equation (26) to calculate the score and the average score is calculated as follows:

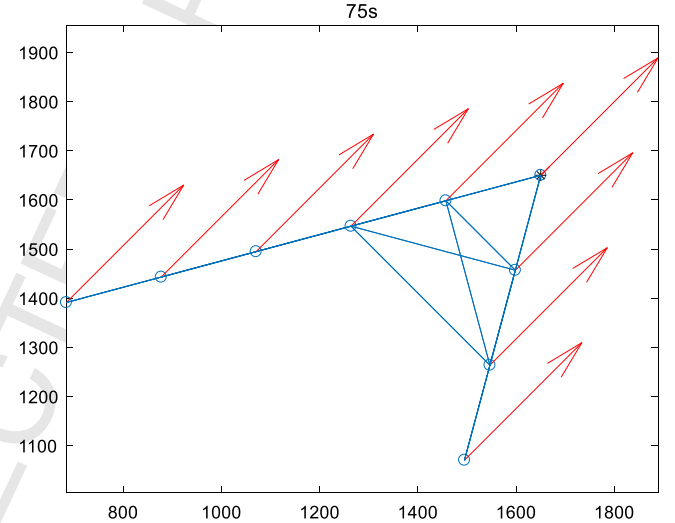
$$\text{score} = \frac{1}{N} \sum_{i=1}^N c_{ij} \quad (31)$$

The results are shown in Fig. 12 and Fig. 13.

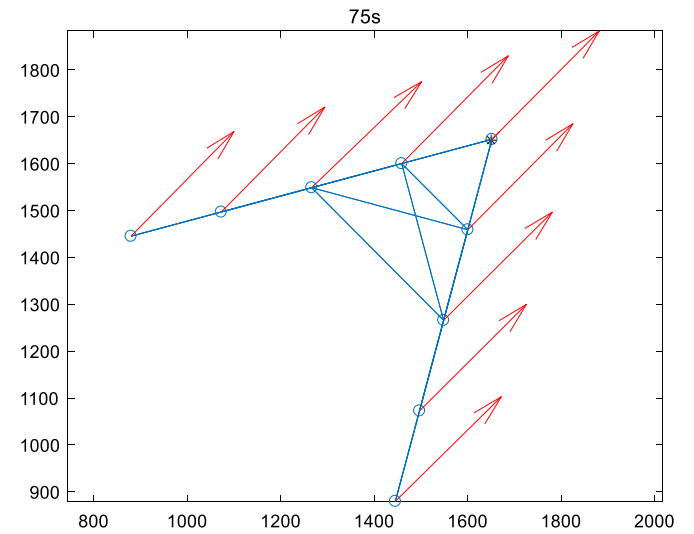
As is shown in Fig. 12 and Fig. 13, with the increase of the UAV's number, the average iteration number using CBBA is lower than the pre-assignment method, which reflects the rapidness of the generation of the desired formation using CBBA. In the meanwhile, the average score of a single UAV is higher, which means the average distance cost is less using CBBA. The allocation method based on CBBA can quickly find an assignment scheme with the lowest total distance cost according to the speed of the UAVs and the positional relationship between the UAVs and the virtual leader



(a) flight with straight-shape



(b) transform by pre-assignment assignment



(c) transform by CBBA assignment

Fig. 14. UAV swarm formation transformation under damage.

at the moment of formation transformation. As a result, the generation and reconstruction of the swarm formation can be realized rapidly and efficiently.

4.4.2. UAV swarm formation transformation under damage

In order to verify the availability of CBBA to allocate positions, the experiment of the formation transformation under damage is simulated. Assume there are 11 UAVs for swarm flight, UAV communication radius $r = 450$, and other parameters are consistent with the above experiment. The UAV swarm needs to change the formation from a straight-shape to a V-shape during performing a task, but two UAVs are damaged (the red circles in Fig. 14(a)) before transforming to V-shape. If the swarm still transform its formation according to the pre-assignment of target positions, it will eventually fail and cannot transform into a reasonable formation. While the proposed formation transformation method with CBBA can solve this problem well. The simulation result under the method of pre-assignment is shown in Fig. 14(b) and the result using CBBA assignment is shown in Fig. 14(c).

It can be seen from Fig. 14 that the formation transformation under the pre-assignment method cannot form a complete V-shape formation, while using CBBA for position allocation eventually forms the desired V-shape formation. The reason is the pre-assignment of the formation transformation cannot adjust its allocation scheme in real time according to the current situation, so the formation transformation is still performed according to the preset allocation scheme. However, the formation transformation using CBBA is a real-time allocation process. Before transforming to the desired formation, the number of target positions can be adjusted in time according to the number of UAVs maintaining communication currently, so that a reasonable allocation scheme can be obtained and the desired formation can be transformed finally.

5. Conclusion

This paper mainly studies a formation maintenance and reconstruction method of UAV swarm based on distributed control. Firstly, the maintenance and reconstruction of the swarm formation are studied based on the consensus algorithm. By designing the control law to control the relative positional relationship between each UAV and the virtual leader to a preset value and control the swarm speed to the desired speed, so as to achieve the generation and maintenance of the desired formation shape. By changing the relative positional relationship between each UAV and the virtual leader, the transformation of the formation is realized under this control law. Then, a new obstacle prediction mechanism is designed to determine whether each UAV needs to avoid obstacles during the flight, thereby reducing the distance traveled by obstacle avoidance. Under the action of the proposed cooperative control law, the swarm can avoid obstacles during the flight and realize the maintenance and reconstruction of the formation effectively. At last, an optimization strategy for target position allocation before the formation transformation based on CBBA is proposed. Each UAV competes its target position according to its

local bid, thereby realizing the rapid position allocation of the target formation in a real-time distributed manner.

The next research direction is to consider the time delay into the communication of the swarm, and study the stability of swarm flight with different time-varying delays.

Declaration of competing interest

The authors declare that the funding which support this research do not lead to any conflict of interests regarding the publication of this manuscript. The authors declare that there is no conflict of interest regarding the publication of this paper.

Uncited references

[12] [13] [14] [15] [17] [18]

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